34870 Electroacoustics – Loudspeaker project

Lab day: building and measuring the crossover (Task 3)

Tuesday 10 November 2026 · Group 10 (Louis, Sophie, Mads) · System D

13:00–15:00 anechoic chamber (Lab E setup, upgraded) · 15:00–17:00 Zi/Nfield station (Lab D setup), building 354

Purpose:

1) Build the passive crossover board (midrange low-pass, tweeter high-pass with L-pad) and set up the MiniDSP digital crossover (woofer low-pass, mid/tweeter high-pass, LR4 at 325 Hz).
2) Measure the finished 3-way system in the anechoic chamber, unit by unit and all together, at 0° and 30°, and check the polarity/phase alignment with two deliberate polarity flips.
3) Measure, at the Lab D station, the impedance the amplifier really sees through the board, check the woofer box tuning (one vent, $L = 160$ mm), and repeat the near-field responses for the low end that the chamber cannot measure.

There is no official lab description for these two sessions. This guide is put together from the Project Guide (Task 3, pages 4–5, and Appendix 1), the Lab D and Lab E descriptions (the two stations), our design (`Project/Final design/design/design.json`) and the lab-day kit in the labs repository (`Project/Final design/lab-day/`). Every MATLAB step below is a section in one of two run files, so the whole day is "click in a section, press *Run Section*, read the printout".

Warnings (from the Lab D/E descriptions)

- The absorbing wedges in the anechoic chamber are fragile and expensive: do not touch them when you carry the boxes in and out. Do not drop screws or small parts through the net.
- Do not touch the drivers, especially the tweeter. Carry the boxes by the cabinet.
- The mid/tweeter box vent stays closed all day. The woofer vents are **both plugged** in the chamber and set to **one vent open at** 160 mm at the Zi/Nfield station.
- System D is shared with Group 4: check nobody needs it in parallel before you move it.
- Do not change the wiring of the MiniDSP in the chamber setup (Project Guide 3, Task 3). Mute and invert are done in the MiniDSP app, never by re-cabling the MiniDSP.

1 Preparation (the evening before)

- ☐ USB stick folder `34870_Group10_ProjectLab_2026-11-10/`: `Labs/` (a mirror of the labs repository, so the relative paths in the scripts work), `Docs/` (Lab D, Lab E and Project Guide PDFs, the three expected-curve plots), `START_HERE_lab-day_README.md` (the checklist) and this guide. The lab PCs have no internet.
- ☐ Offline tests pass on a laptop: `test_project` and `test_zinf` both print `PASS` (checked 9 Nov).
- ☐ Print this guide (or have it on a laptop). Pages 12 and 13 (expected curves) are the ones to have next to the screen.
- ☐ Bring: tape measure, multimeter, a pen, a phone for photos. Ask for an LCR meter in the lab.
- ☐ Vicente (vcuhe@dtu.dk, room 016, b. 352) is booked for 13:00 (his announcement of 27 Oct: every group gets a short briefing on the new equipment the first time it uses it).

2 The day at a glance

Time	Where	What	File / section
12:45	outside 028/025	meet, check this guide, the stick and the parts from the course baskets	–
13:00	chamber control room	Vicente's briefing on the MiniDSP and the filter panels (booked); ask the open questions in Section 6	–
13:15	control room + chamber	three jobs in parallel: build the board (A), set the MiniDSP (B), place the system and the microphone (C)	Sec. 4.2, 4.3, 4.4
13:35	lab PC	copy the files, start MATLAB, `project_dims`, `test_project` = PASS	`project_run.m` §0
13:40	lab PC	scope check of the MiniDSP at 325 Hz; one-state trial run to set the amplifier volume, then freeze it	Sec. 4.6
13:50	lab PC	runs R1–R5, about 4.5 min each plus a quick look at the printout	`project_run.m` R1–R5
14:40	lab PC	spare time: repeat a bad run, or R6 with a changed Out 1 gain	§6
14:50	control room	save the MiniDSP config, photograph the board, copy data, clean up, carry the system to the Lab D station	§7
15:00	Zi/Nfield (026)	connection board with the 33 Ω resistor, multimeter values, `test_zinf` = PASS	`zinf_run.m` §0
15:15	Zi/Nfield	board impedance Z1, Z2, Z3	Z1–Z3
15:40	Zi/Nfield	woofer impedance in the box at the final tuning, Z4	Z4
16:00	Zi/Nfield	near field cone and vent, NF1, NF2 (resistor shorted, Nexus)	NF1, NF2
16:30	Zi/Nfield	if time: both vents open (never measured before)	X
16:45	Zi/Nfield	copy data, delete our files, plugs back, return parts and cables	last section

Each run is about 4.5 min in the chamber (three states of 64 periods of 1 s each, plus two 30 s beep breaks) and under a minute at the Lab D station. The plan has about 25 minutes of slack in each slot; use it to repeat a run rather than to skip the checks.

3 Equipment

Anechoic chamber (upgraded Lab E setup):

- Desktop PC with internal sound card (line-out looped to line-in and sent on to the MiniDSP)
- UMIK microphone (MiniDSP) with the long USB cable
- MiniDSP box + its control app on a dedicated PC
- Stereo audio amplifier (ch. 1 = woofer, ch. 2 = passive board)
- Passive filter panel (one of 10) with screw terminals
- Course baskets with coils, capacitors, resistors, banana cables
- System D: woofer box (Scan-Speak 26W/8534G00) + white mid/tweeter box (DALI Spektor 2, S/N 7446912 – L), two vent plugs
- Oscilloscope (for the MiniDSP check)

Zi/Nfield station (Lab D setup):

- Desktop PC with NI-4431 card (AO0 out, AI0 and AI1 in, ±10 V)
- Audio amplifier
- Circuit connection board with the 33 Ω series resistor
- Multimeter
- Nexus microphone amplifier + B&K or GRAS microphone
- Our passive board from the chamber, and System D

Ours: USB stick, this guide, tape measure, multimeter, phone.

4 Part 1 – Anechoic chamber, 13:00–15:00

4.1 Signal chain

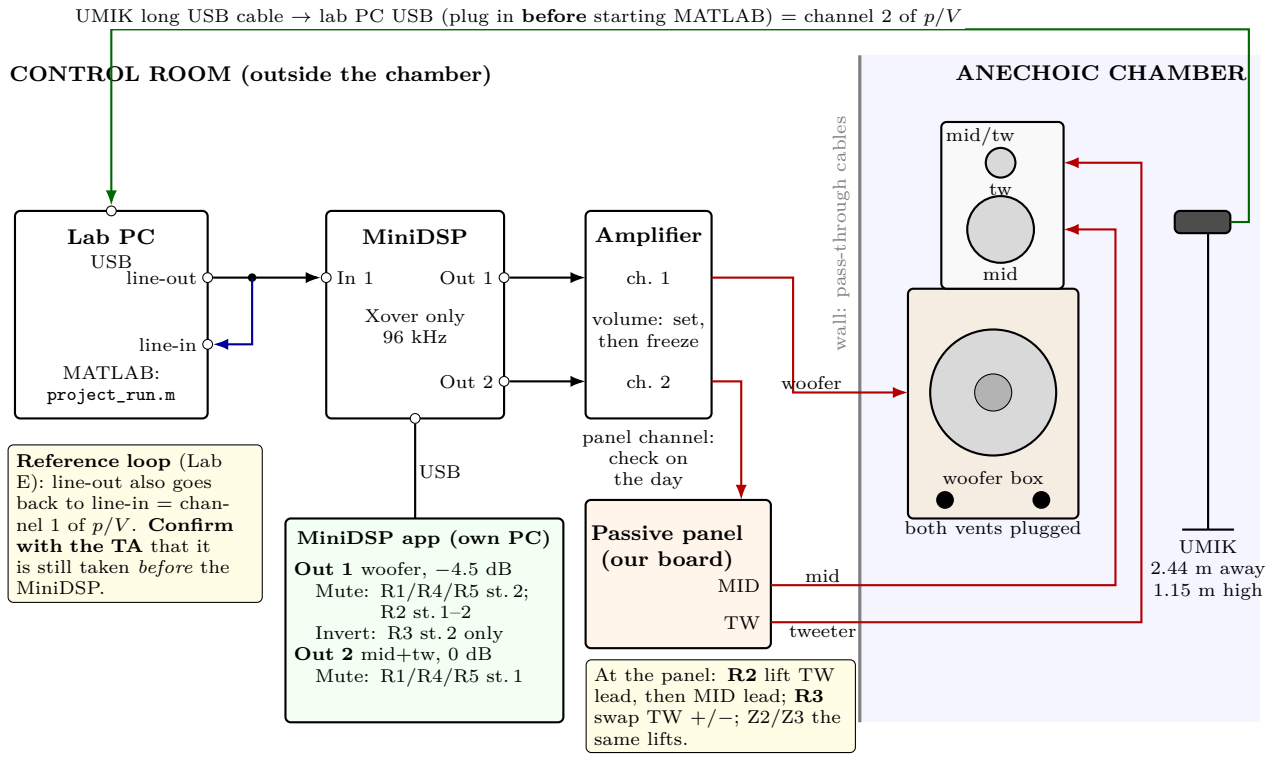


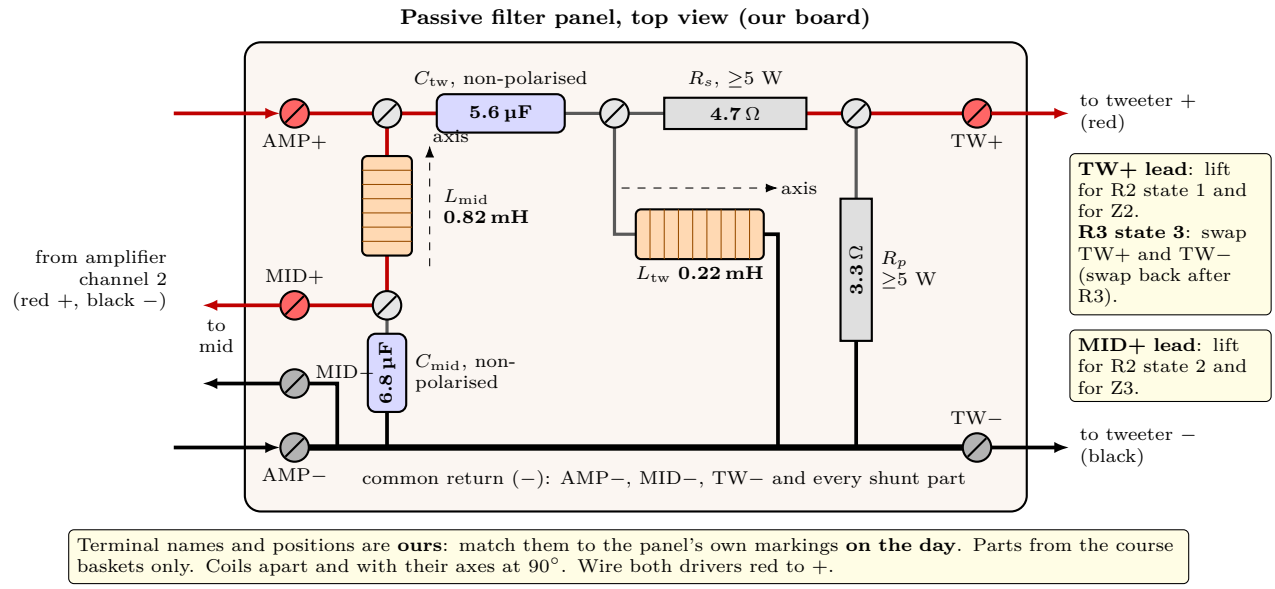
Figure 1: Wiring in the chamber session. The script computes $H = p/V$ from the UMIK (channel 2) and the line-in reference (channel 1), as in Lab E. Everything except the boxes and the microphone stays in the control room, including the passive panel, so muting and lead lifting never need the chamber door. The Project Guide says one amplifier channel feeds the panel and the other goes straight to the chamber: check on the day which one, and make sure MiniDSP Output 2 drives the panel channel.

The one thing to confirm with Vicente before measuring

Our scripts assume the line-in reference is still taken from the sound card line-out *before* the MiniDSP, as in Lab E. Then H is the response of the whole system (MiniDSP + amplifier + board + drivers), which is what the model curves show. If the reference is now tapped after the MiniDSP or after the amplifier, the measured curves are not the system response: stop and adapt the plan with him.

4.2 The passive board (person A, 13:15–13:35)

The board sits on amplifier channel 2. Both branches start at the amplifier + terminal; every shunt part goes to the common return (−). Wire both drivers red to +.



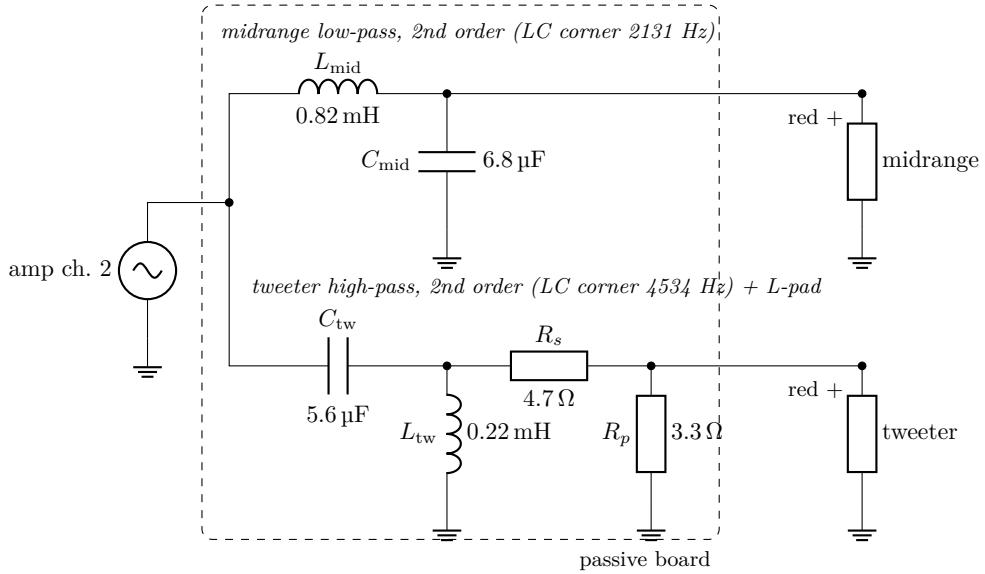


Figure 2: The passive board. Top: how to wire it (a top-view sketch; the terminal names are ours, match them to the panel on the day). Bottom: the same circuit as a schematic. All values are single parts from Appendix 1, Table 1 of the Project Guide. Acoustic crossovers in the model: woofer/midrange 324 Hz (MiniDSP), midrange/tweeter 3.76 kHz (this board). The L-pad lowers the tweeter by about 11 dB.

Part	Value	Where it goes	Check the basket part
L_{mid}	0.82 mH	series, amp + → midrange +	carries up to ∼1.9 A rms at 10 W
C_{mid}	6.8 µF	across the midrange (+ to −)	**non-polarised** (film or bipolar), ≥25 V
C_{tw}	5.6 µF	series, amp + → tweeter node	**non-polarised**, ≥25 V
L_{tw}	0.22 mH	tweeter node → return	–
R_s	4.7 Ω	series, node → tweeter +	**power resistor,** ≥5 W
R_p	3.3 Ω	across the tweeter	power resistor, ≥5 W

☐ **Where the parts come from:** only the course baskets (Project Guide 3.2), values from Appendix 1, Table 1. There are at least two parts of each value. Do not combine parts to make an in-between value; all six of our values are in the table. Return unused parts and every banana cable to their basket afterwards, and report parts that are running low, missing or faulty.

☐ Measure each part with the LCR meter if the lab has one, and each coil's DC resistance with the multimeter. Write the numbers into `project_dims.m` (`d.board`, `d.coil_dcr`). For reference: coil resistance of

0.3/0.6/1.0 Ω per mH lowers the summed response by at most 0.23/0.45/0.74 dB, and 400 random boards within the part tolerances stayed inside a 1.7 dB band.

☐ Keep the two coils apart and at right angles to each other (so one does not pick up the other's field).

☐ Note which panel (1–10) we use: `d.panel_no`. The load the board puts on amplifier channel 2 never drops below 4.4 Ω (model).

4.3 The MiniDSP (person B, 13:15–13:35)

Steps from the Project Guide, section 3.1, with our values:

1. Open the MiniDSP software on its PC and click *Connect*.
2. **Routing matrix:** only *Input 1* → *Output 1* and *Input 1* → *Output 2* active.
3. **Outputs:** only outputs 1 and 2 are used. On both, only *Xover* is active: PEQ, Comp and FIR must show as bypassed (we cannot reproduce them in LTspice).
4. **Output 1 (woofer):** Xover → *Advanced*, delete the contents of the text window, paste the whole of `Project/Final design/data/dsp/MiniDSP_woofer_out1.txt` (open it in Notepad on the stick and copy from there, never from this PDF: hidden characters), click *Process*. It is two identical Butterworth low-pass biquads at 325 Hz = one LR4 low-pass.
5. **Output 2 (mid + tweeter):** same with `MiniDSP_midtw_out2.txt` (the LR4 high-pass). Synchronise the configuration with the DSP card if the app asks.
6. **Gains:** Output 1 −4.5 dB, Output 2 0 dB. Invert off on both. Delay 0 on both (the "$d = -1.15$ m" in our report is only the distance correction for drawing phase plots, not a delay to type in).
7. **Sample rate:** the coefficients only give 325 Hz at 96 kHz. Check the plugin says 96 kHz (`d.minidsp_fs`).
8. Save the configuration (slot and file); write the names in `d.minidsp_cfg`.

Setting	Value
Routing	Input 1 → Output 1, Input 1 → Output 2, nothing else
PEQ, Comp, FIR	bypassed on both outputs
Output 1 Xover, Advanced	`MiniDSP_woofer_out1.txt` (LR4 low-pass 325 Hz), Process
Output 2 Xover, Advanced	`MiniDSP_midtw_out2.txt` (LR4 high-pass 325 Hz), Process
Gain	Output 1 −4.5 dB, Output 2 0 dB
Invert / polarity	off on both (R3 ticks Out 1 for one state, then untick)
Delay	0 on both
Sample rate	96 kHz

Table 1: MiniDSP settings. The coefficient texts are in Appendix C.

4.4 Placing the system and the microphone (person C, 13:15–13:35)

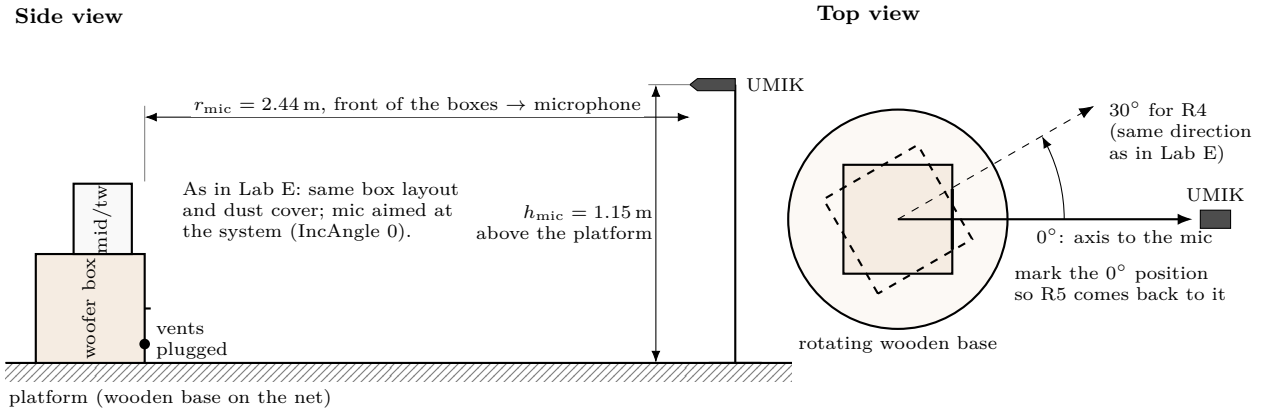


Figure 3: Placement in the chamber, as in Lab E (side view roughly to scale). The microphone stays put all session; only the wooden base turns for R4 and back for R5.

- ☐ Same placement as in Lab E, because the model curves are built from the Lab E data: mid/tweeter box on the woofer box in the Lab E position, microphone 2.44 m from the front of the boxes and 1.15 m above the platform, pointing at the system (IncAngle 0). Write the real numbers into `d.r_mic`, `d.h_mic`, `d.layout`.
- ☐ Both woofer vents plugged ("you may just cover the vents", Task 3): the chamber is not valid below 125 Hz, so the vent tuning is measured at the Lab D station instead. Filling stays in.
- ☐ Dust cover as in Lab E (`d.dust_cover`). Read the UMIK serial on its label (0329, 0332 or 0335; Lab E was 708-0332) and put it in `d.UmikSN`. Note the chamber temperature.
- ☐ Connect the woofer to amplifier channel 1 and the two board outputs to the midrange and the tweeter, red to +. Mark the 0° position of the wooden base so R5 comes back to it.
- ☐ Plug the UMIK USB cable into the PC **before** MATLAB is started, or the routine does not find it.

4.5 Files on the lab PC (13:35)

1. Copy the folder `Labs/` from the stick to the lab PC (for example `Desktop\Group 10\Labs`). Copy the whole folder: the scripts find the course routines and the Lab D reference data by relative paths (`Labs/Lab E/matlab/course`, `Labs/Lab D/matlab`, `Labs/Lab D/data`).
2. Start MATLAB (UMIK already plugged in), `cd` to `Labs/Project/Final design/lab-day/matlab` and open `project_run.m`.
3. Click in section **0** and press *Run Section* (Ctrl+Enter). It opens `project_dims.m`: fill in everything you have (Appendix B) and save. Then it runs `test_project`, which simulates a whole run without hardware.

What the output should say

`test_project` ends with `PASS`. If it prints `FAIL` or an error, the copy is incomplete: copy `Labs/` again. New data will be saved in `lab-day/data/` as `project_<tag>.mat` (a repeated tag gets `_2`, `_3`; nothing is ever overwritten).

4.6 Electrical check and level setting (13:40)

Scope check of the MiniDSP. Play a 325 Hz sine from the lab PC, e.g. in MATLAB:

```
fs = 48000; t = (0:10*fs-1)/fs; sound(0.1*sin(2*pi*325*t), fs)   % 10 s at 325 Hz
```

and look at the MiniDSP input and its two outputs on the scope (amplifier volume low).

What the scope should show

Output 1 = input −10.5 dB (factor 0.30: −6 dB from the filter at its corner, −4.5 dB gain). Output 2 = input −6.0 dB (factor 0.50). Output 1 and Output 2 **in phase** (an LR4 pair is 360° apart at the crossover). The ratio Out 1/Out 2 = −4.5 dB is the robust number, even if the input gain is not 0 dB.

Trial run for the amplifier volume. Start with the volume low and run once, outside `project_run.m`, with one state (about a minute):

```
m = measure_project('trial_0', 0, {'all'}, 'note', 'level trial');
```

Raise the volume until the UMIK peak printed after the run is well above 0.01 and below 0.95 (no warnings), and the measured curve is smooth up to 20 kHz. Then write the knob position in `d.amp_volume` and **do not touch the volume or the microphone again** until R5 is done.

4.7 Procedure: the five runs (13:50–14:40)

One run measures one to three states in one go. Between two states the routine plays about 30 s of beeps: that is when you switch. States inside one run share a phase reference; separate runs do not (the UMIK has its own USB clock), so phases are only compared within a run. Muting happens in the MiniDSP app; lead swapping at the board, outside the chamber. After every run the script saves the file, prints the peaks, plots the measured curves on top of the model (dashed, shifted by a fitted level) and prints the checks below.

Run	Angle	States (switch during the beeps)	What it proves
R1	0°	woofer → mid+tw → all	each channel against the model; channel balance; woofer + mid/tw = all
R2	0°	mid → tw → all	the passive board, driver by driver
R3	0°	all → woofer inverted → tweeter swapped	phase alignment: notches at 329 Hz and 3.9 kHz
R4	30°	woofer → mid+tw → all	off-axis, against the Lab E 30° model
R5	0°	woofer → mid+tw → all	repeat of R1: drift and placement

R1 – on axis, woofer / mid+tweeter / all

`project_run.m`, section R1

Before: Output 2 **muted** in the app. **Beeps 1:** unmute Output 2, mute Output 1. **Beeps 2:** unmute Output 1.
Runs `measure_project('R1_split_0', 0, {'woofer','midtw','all'}, 'note','reference run')`.

What the output should say

```
R1_split_0  (0 deg)
  woofer  level offset vs model   +x.x dB, shape error (rms, 1/6 oct) 1.x dB
  midtw   level offset vs model   +x.x dB, shape error (rms, 1/6 oct) 1.x dB
  all     level offset vs model   +x.x dB, shape error (rms, 1/6 oct) 1.x dB
  woofer - midtw offset: +0.x dB (want within +-1 dB; else adjust the Out 1 gain by ...)
  sum check |woofer+midtw| vs |all|: worst 0.xx dB (1/6 oct, 150 Hz-15 kHz)
```

- The **level offsets** themselves can be anything (amplifier volume, MiniDSP input gain). What matters is that **woofer and midtw agree within about** 1 dB. If not, the two amplifier channels differ: change the Output 1 gain by the printed amount, note the new value in `d.minidsp_out1_gain`, and repeat R1 (the tag gets `_2`).
- **Shape error** about 1–2 dB or less. **Sum check** under about 0.5 dB; more means something moved or changed between the states: repeat the run.
- The plot should look like the top panel of Figure 5: woofer and mid+tweeter cross near 325 Hz, the sum is within ±2 dB from 150 Hz to 15 kHz (4.0 dB spread, 1/6 octave).

R2 – on axis, midrange / tweeter / all

section R2

Before: Output 1 **muted**, **tweeter** lead off at the board. **Beeps 1:** tweeter lead on, midrange lead off. **Beeps 2:** midrange lead back on, unmute Output 1.
Runs `measure_project('R2_units_0', 0, {'mid','tw','all'})`.

What the output should say

Each driver's shape error is small (it is judged only where that driver is loud: midrange 600 Hz–2.5 kHz, tweeter 5–15 kHz). The midrange falls off above about 4 kHz and the tweeter takes over around 3.8 kHz (Figure 5, top). If one driver is far off while the other is fine, the fault is in that branch of the board (a part in the wrong place, a polarised capacitor, a loose screw).

R3 – polarity checks: all / woofer inverted / tweeter swapped

section R3

Before: everything normal. **Beeps 1:** tick **Invert** on Output 1. **Beeps 2:** untick it, and swap the tweeter's + and − at the board.
Runs `measure_project('R3_flip_0', 0, {'all','flipW','flipT'})`.

What the output should say

```
  flipW: notch 3xx Hz, -2x.x dB below the normal sum      (model: 329 Hz, -20.9 dB)
  flipT: notch 3xxx Hz, -1x.x dB below the normal sum     (model: 3928 Hz, -14.2 dB)
```

A deep notch proves the two drivers are in phase at the crossover with normal wiring. A shallow notch or one in the wrong place means a polarity or phase error: check red to + on both drivers, and that the Invert box really was ticked. Compare with Figure 5, bottom panel.

Afterwards: untick Invert (if still ticked) and swap the tweeter back. Check both before R4.

R4 – 30°, woofer / mid+tweeter / all

section R4

Turn the wooden base 30°, in the same direction as in Lab E. Same switching as R1.
Runs `measure_project('R4_split_30', 30, {'woofer','midtw','all'})` and compares with the 30°

model (Figure 6).

What the output should say

Same checks as R1. Woofer – midtw offset close to R1's value. Expected spread of the sum: about 6.3 dB (150 Hz–15 kHz, 1/6 octave); the top end rolls off above 10 kHz (tweeter directivity).

R5 – back to 0°: repeat of R1 section R5

Turn back to the 0° mark. Tweeter polarity normal, no invert. Same switching as R1.
Runs `measure_project('R5_repeat_0', 0, {'woofer','midtw','all'}, 'note','repeat of R1')` and compares the "all" curve with R1.

What the output should say

```
R5 vs R1, all three: median +0.0x dB, worst 0.x dB
```

Median within a few tenths of a dB. A bigger shift means the system, the microphone or the volume moved during the session; write it down (it is the uncertainty of the whole series).

Section 6 (optional, 14:40). If R1 asked for a different Output 1 gain, or another change is tried, run it with a new tag and a note saying exactly what changed, e.g.

```
m = measure_project('R6_gain_0', 0, {'woofer','midtw','all'}, 'note', 'Out 1 gain -3.5 dB');
```

4.8 Before leaving the chamber (14:50)

- ☐ Save the MiniDSP configuration (slot and file); copy the file to the stick. The listening test and the demo start from it.
- ☐ Photograph the board from above, with the part values readable (report).
- ☐ Copy `Labs/Project/Final design/lab-day/data/` to the stick, into the same path. Check that the files open on the stick.
- ☐ Untick Invert, tweeter wired normally, vent plugs still in.
- ☐ Return unused parts and banana cables to their baskets. Take the board and the system to the Lab D station (careful with the wedges and the drivers).

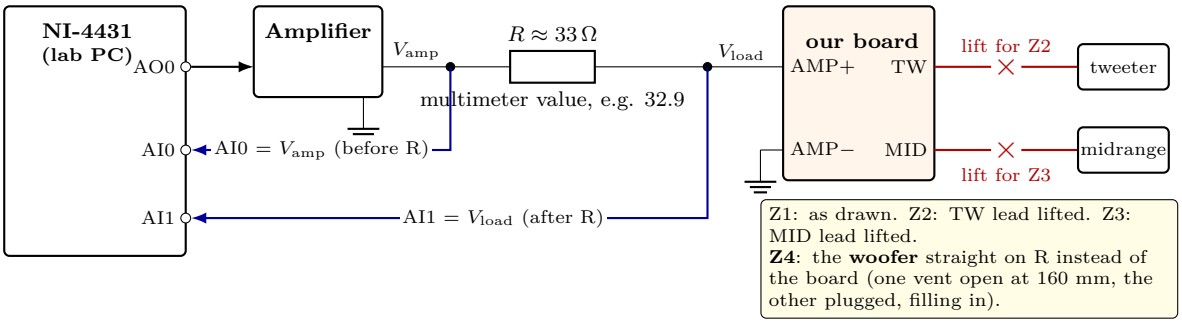
5 Part 2 – Zi/Nfield station (Lab D setup), 15:00–17:00

This is the Lab D setup, not the MiniDSP chain: the NI card's AO0 drives the amplifier, the amplifier drives the load through the series resistor on the circuit connection board, AI0 reads the amplifier voltage and AI1 the load voltage (for impedance) or the Nexus output (for near field). The impedance is

$$Z_L = R\,\frac{V_{\text{load}}/V_{\text{amp}}}{1 - V_{\text{load}}/V_{\text{amp}}},$$

computed by `zinf_calc.m`. Two Lab D lessons are built into the scripts: R must be the **multimeter value of the series resistor** (about 32.9 Ω, never R_E), and when AI0 sits on a DC offset of several volts (the amplifier's output stage; in Lab D it read half the amplifier voltage), the script uses $V_{\text{amp}} = 2 \cdot \text{AI0}$ and says so.

(a) Impedance, Z1–Z4: series resistor in



(b) Near field, NF1/NF2: resistor shorted, microphone via the Nexus

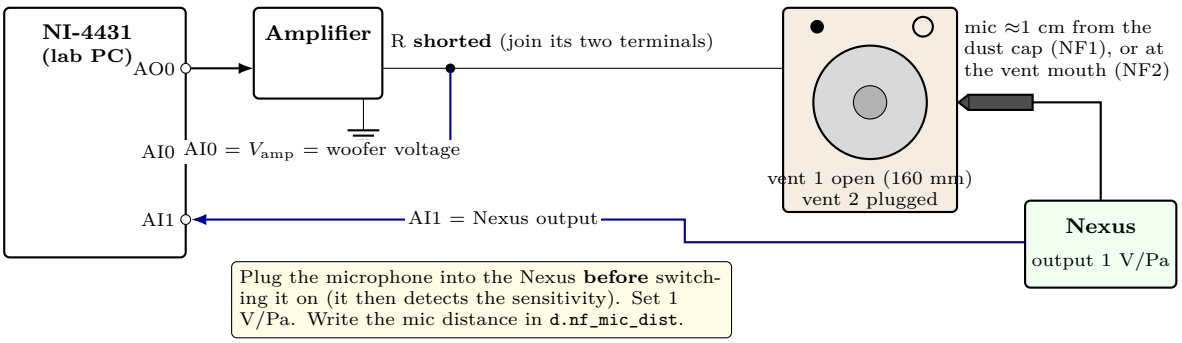


Figure 4: Wiring at the Zi/Nfield station (channels as in the Lab D description and the course routine `meas_mag_spec2_LabD.m`: AO0 drives the amplifier, AI0 reads the amplifier output, AI1 the load voltage or the Nexus). The NI inputs share ground, which is why the resistor voltage is never measured directly. Which terminals of the connection board are which: check on the day (as in Lab D).

5.1 Setup and files (15:00–15:15)

1. Copy `Labs/` (now with the chamber data in it) to this PC as in Section 4.5. Start MATLAB, go to `lab-day/matlab` and open `zinf_run.m`.
2. Fit the 33 Ω resistor in the connection board. Measure it with the multimeter and type the value in section 0 (`R = 32.9;`, say), and in `d.R_series`.
3. With the multimeter, also measure: the board input with both drivers connected (`d.Rdc_board`: expect about 5 Ω, the midrange R_E plus L_{mid}'s resistance; the tweeter branch is open at DC), the coil resistances if not done yet, and both vents' inner diameters and the second vent's telescope range (`d.vent1`, `d.vent2`).
4. Run section 0: it opens `project_dims.m` (fill in, save) and runs `test_zinf`.

What the output should say

`test_zinf` prints `PASS` (it simulates the board, the woofer at 160 mm and the near field, with the Lab D half-voltage AI0). Data are saved as `lab-day/data/zinf_<tag>.mat`.

Level. Start with the amplifier volume low. After each run the script prints `rms AI0 ..., AI1 ... | peak ... | AI0 DC ... -> k = 1 or 2`. As in Lab D, aim for about 0.1 V rms on the load voltage; there must be no warning about the ±10 V rail. A `k = 2` message is expected if the amplifier output sits on a DC offset, as in Lab D.

5.2 Procedure

Run	Section / tag	Setup	Expected
Z1	`board`	amp → R → board input, both drivers on	$\|Z\|$ min 4.4 Ω at ∼5.9 kHz; 6.4 Ω at 50 Hz; peak ∼19 Ω near 90 Hz
Z2	`board_mid`	tweeter lead off at the board	min 4.8 Ω at ∼200 Hz, rising to ∼100 Ω at 20 kHz
Z3	`board_tw`	midrange lead off at the board	∼570 Ω at 50 Hz falling (capacitor), min 4.2 Ω at ∼5.3 kHz
Z4	`woofer_L160`	woofer straight on R; one vent open at 160 mm, other plugged, filling in	$f_b \approx 23.4$ Hz (Lab D), peaks f_L 12.0 and f_H 45.5 Hz; model f_B 24.6
NF1	`nf_cone_L160`	near field, mic ∼1 cm from the dust cap; R shorted	Lab D: peak 13.7 dB re 1 Pa/V at 56 Hz, dip at ∼22 Hz
NF2	`nf_vent_L160`	near field, mic at the vent mouth	Lab D: peak 13.2 dB re 1 Pa/V at 29 Hz
X	`woofer_2vents`	both vents open (optional)	never measured; quiz 1 only estimated it

Z1–Z3 – the board as the amplifier sees it

`zinf_run.m`, sections Z1, Z2, Z3

Connect amplifier → series resistor → board input. Z1 with both drivers, Z2 with the tweeter lead off, Z3 with the midrange lead off (put each lead back afterwards). The tag decides which model curve is used (`mid` → Zmid, `tw` → Ztw, otherwise Zboth). About 20 s each (20 Hz–24 kHz, 0.125 Hz resolution).

What the output should say

```
board vs model Zboth (50 Hz-19.7 kHz): median +0.xx dB, worst x.xx dB around that
  |Z| min 4.xx ohm at 5xxx Hz   (model 4.40 ohm at 5880 Hz)
```

Median within about 1.5 dB (a warning is printed if not: check R, the k message and the leads). $|Z|$ at the low end close to the multimeter value `d.Rdc_board`: that checks R and k at the same time. A minimum below 3.5 Ω triggers a warning: look for a short or both branches on one driver. The plot shows the measurement against the model with ideal coils (dashed) and with 0.6 Ω/mH coils (dotted); see Figure 7.

Z4 – woofer in its box at the final tuning

section Z4

Connect the woofer directly to the resistor output (no board). Vent 1 open at 160 mm, vent 2 plugged, filling in, as designed. Range 1 Hz–10 kHz.

What the output should say

```
woofer_L160: fL 12.0 Hz, fb 23.4 Hz, fH 45.5 Hz, |Z| at 1.0 Hz x.xx ohm
  Lab D labD_vent_L160.mat: fL 12.0, fb 23.4, fH 45.5 Hz   (design model fB 24.6 Hz)
```

f_b within about 1 Hz of Lab D's 23.4 Hz, two peaks either side, and $|Z|$ at 1 Hz near the woofer's R_E. If f_b is far off: check the vent length, that the other vent is really plugged, and the filling.

NF1, NF2 – near field at the final tuning

sections NF1, NF2

Short the series resistor (join the two terminals). Connect the microphone to the Nexus **before** switching the Nexus on (it then detects the microphone's sensitivity), set the output to 1 V/Pa, Nexus output to AI1. Microphone about 1 cm from the dust cap (NF1), then at the vent mouth (NF2); write the distance in `d.nf_mic_dist`. `Nav = 4` (about 40 s).

What the output should say

```
nf_cone_L160: |p/V| peak 13.x dB re 1 Pa/V at 5x.x Hz (10-200 Hz)
  vs Lab D labD_nf_cone_L160.mat (10-200 Hz): median +0.xx dB, worst x.xx dB around that
```

The new curve lies on top of Lab D's 160 mm curve (median within about 1 dB; the exact mic distance moves the level). The cone shows its dip near f_b, the vent its peak near f_b. If the level is far off: Nexus not at 1 V/Pa, microphone plugged in after the Nexus was switched on, or the resistor not shorted.

X (optional, 16:30): both vents open. Measure the second vent's diameter and length first, then uncomment the two lines in section X and fill in the lengths in the note. This has never been measured: quiz part 1 only estimated it (vent mass $M_{AP} \approx 52$), so it is a nice extra for the report.

5.3 Before leaving (16:45)

- ☐ Copy `lab-day/data/` to the stick (now chamber + Zi/Nfield files). Check they open.
- ☐ Delete our files from **both** lab PCs (the copied `Labs/` folders).
- ☐ Put the vent plugs back the way the box is meant to be stored ("all boxes come with the vents closed by plugs"). Remove the short from the connection board.
- ☐ Keep the board (if Vicente says we may) for the listening test and the demo; return everything else.

6 Open questions for Vicente (13:00 briefing)

1. Is the line-in reference still tapped from the sound card line-out *before* the MiniDSP? (Our p/V and the model assume it is.)
2. Which MiniDSP model is it, and does the plugin run at 96 kHz? (Our coefficients only hold at 96 kHz; at 48 kHz the crossover would sit at 162 Hz.)
3. Are the two amplifier channels set to the same gain? (Our -4.5 dB woofer gain assumes so; R1 checks it.)
4. Is there an LCR meter? What are the voltage and power ratings of the basket capacitors and resistors (non-polarised caps, $\geq$5 W for the 4.7 Ω)?
5. May we use the *Invert* button in the app for the polarity check in R3?
6. Do we keep the board (and the panel) for the listening test on 17 Nov and the demo on 3 Dec?

7 After the lab: what the data is for

- **Report, Task 3:** measured vs. model for R1–R4 (offsets, shape errors, the two notches), the board photo and measured part values, the board impedance against the model, and the R5 repeat as the uncertainty of the series. Put `data/` back into the repository's `lab-day/data/`.
- **Report, Task 1:** Z4 and NF1/NF2 confirm (or update) the box tuning with the final board and filling, and X adds the both-vents case.
- **Listening test (Task 4):** Tue 17 Nov, 13:00, IEC listening room. The room has its own PC, MiniDSP, amplifier and cabling: bring the board and the saved MiniDSP configuration / filter texts. If R1 changed the Output 1 gain, use the new value.
- **Project quiz part 2:** Mon 30 Nov (final results and discussion).
- **Demo:** Thu 3 Dec, room 019: bring only the board and the filter texts; boxes and equipment are there.

8 Expected results (model)

The model is our design (`design.json`) applied to the Lab D impedances and the Lab E responses with the vent plugged (`model_curves.py`). The grey band below 125 Hz is where the chamber is not valid. `compare_project` shifts each model curve by one fitted level before comparing, so read these for shape, not absolute level.

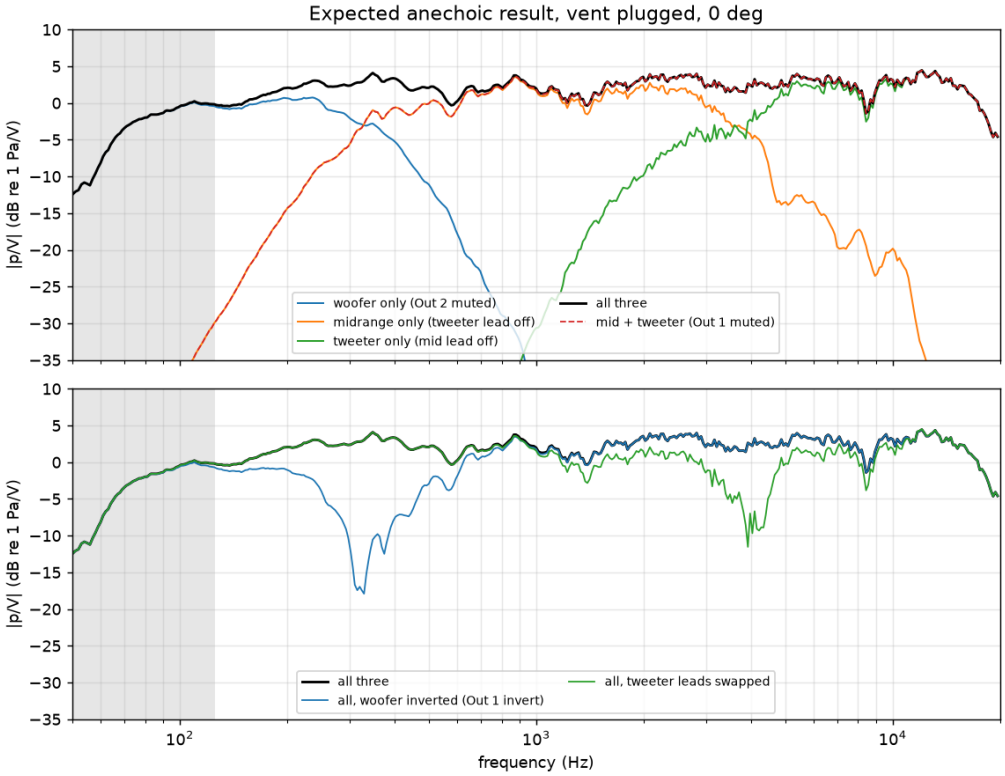


Figure 5: Expected result at 0°. Top: R1 and R2 states (woofer only, mid+tweeter, midrange only, tweeter only, all). Bottom: R3 states (all, woofer inverted, tweeter swapped) with the notches at 329 Hz (−21 dB) and 3.9 kHz (−14 dB).

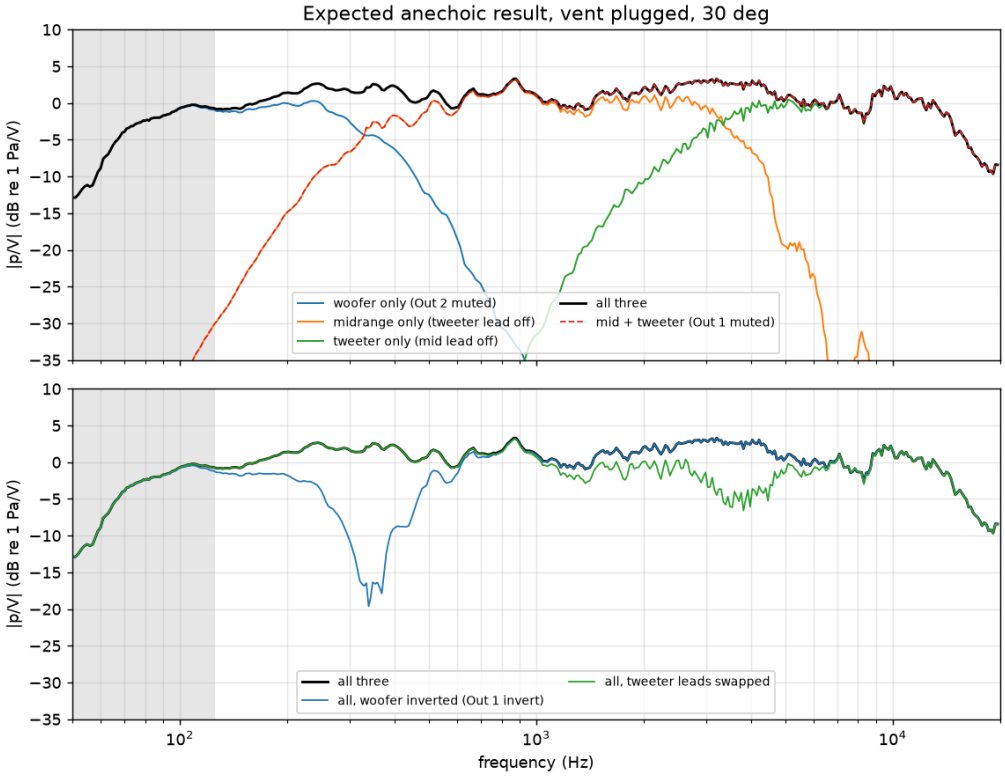


Figure 6: Expected result at 30° (R4). Sum spread 6.3 dB from 150 Hz to 15 kHz.

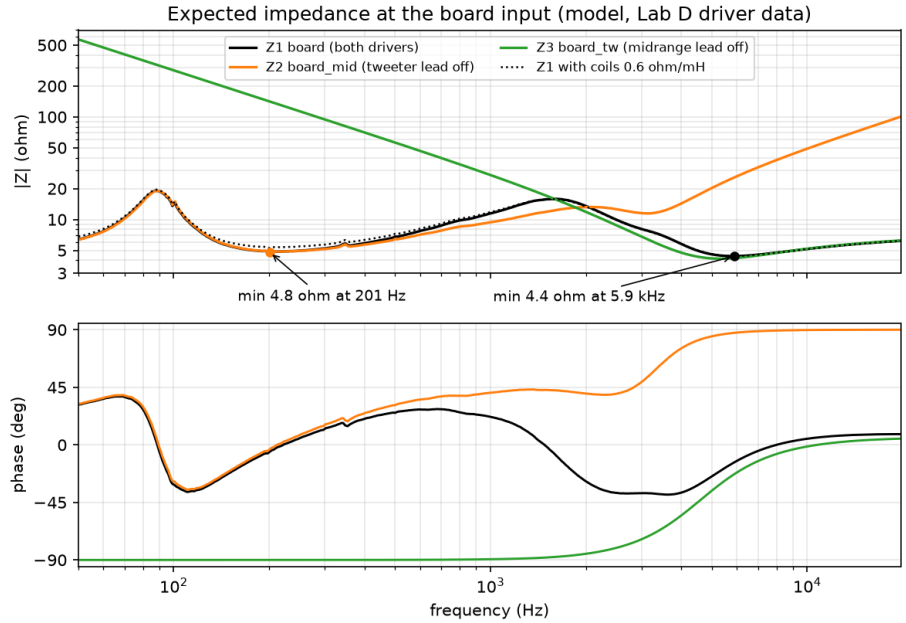


Figure 7: Expected impedance at the board input for Z1–Z3 (log scale; same data as `model/expected_zi.png`). Dotted: coils with 0.6 Ω/mH resistance, which raises the low end by about 0.4 Ω.

A Troubleshooting

Symptom	What to do
"No Umik" / microphone not found	Close MATLAB, check the UMIK USB cable, start MATLAB again (the device is only registered at start-up).
"beeps.wav not found"	`cd` into `Labs/Lab E/matlab/course` and run from there, or copy `Beeps.wav` next to `project_run.m`.
`test_project` / `test_zinf` not PASS	The copy of `Labs/` is incomplete: copy the whole folder again.
"Unknown state(s)"	A typo in a state name. Valid: `woofer midtw mid tw all flipW flipT`.
Warning: time signal near full scale	Turn the amplifier down. Before R1: fine. During the series: redo from R1, since the volume must be the same for all runs.
Warning: UMIK signal very small	A mute left on, a lead off, the amplifier too low, or the UMIK cable.
Woofer – midtw offset more than ±1 dB	Change the Output 1 gain by the printed amount, note it (`d.minidsp_out1_gain`), repeat R1.
Sum check above 0.5 dB	Something changed between states (a lead, a mute, someone in the chamber): repeat the run.
No notch, or in the wrong place, in R3	Polarity: red to + on both drivers? Invert really ticked/unticked? Tweeter swapped in the third state only?
mid+tw far from the model, woofer fine	Board wiring: run R2 to find which branch.
Scope: Out 1 about −29 dB, Out 2 about −0.5 dB at 325 Hz	The MiniDSP runs at 48 kHz: the coefficients put the crossover at 162 Hz. Ask Vicente.
Scope: Out 1 and Out 2 equal	The −4.5 dB gain on Output 1 is not set.
Scope: outputs in opposite phase	An Invert box is ticked, or a filter text went into the wrong output.
"Give the measured series resistor"	Set `R` in `zinf_run.m` section 0 to the multimeter value (20–50 Ω).
Warning: samples on the ±10 V rail	Turn the amplifier down; that run is not usable, repeat it.
Message "AI0 is on a DC offset … k = 2"	Expected (as in Lab D). Check $\|Z\|$ at low frequency against the multimeter.
Board level off by more than 1.5 dB	Wrong R, wrong k, or a lead on the wrong terminal.
Board $\|Z\|$ min below 3.5 Ω	A short, or both branches on the same driver.
Woofer f_b far from 23.4 Hz	Vent length, the second vent not plugged, filling moved.

Symptom	What to do
Near-field level far from Lab D	Nexus not at 1 V/Pa, microphone plugged in after the Nexus was on, resistor not shorted, or a very different mic distance.

B The hand-written logbook: `project_dims.m`

Every data file stores a copy of this function's output, so fill it in before the first run and update it when something changes. NaN or empty means "not noted yet". Lengths in metres.

Field	What to write
`UmikSN`	UMIK serial from the label (0329, 0332 or 0335); Lab E was `'708-0332'`
`T_celsius`	chamber temperature
`r_mic`, `h_mic`	front of the boxes to the microphone; microphone height (Lab E: 2.44, 1.15)
`layout`	where the mid/tweeter box sits on the woofer box
`vents`, `filling`, `dust_cover`	preset: both plugged, filling in; dust cover as in Lab E
`ref_channel`	confirm with Vicente: line-out looped to line-in, before the MiniDSP
`amp_volume`, `amp_balance`	knob position(s); how the two channels were checked equal
`panel_no`	passive filter panel number (1–10)
`minidsp_cfg`	configuration slot / saved file name
`minidsp_in_gain`	input gain in dB
`minidsp_out1_gain`, `..._out2_gain`	−4.5 and 0 dB (update if R1 changes it)
`minidsp_invert`, `minidsp_fs`	"none"; sample rate shown in the plugin (96 kHz)
`board`	measured value of each of the six parts (LCR meter), next to the nominal
`coil_dcr`	DC resistance of L_{mid} and L_{tw} (multimeter)
`R_series`	series resistor at the Zi/Nfield station, multimeter (Lab D: 32.9)
`Rdc_board`	multimeter across the board input, drivers connected (about 5 Ω)
`vent1`, `vent2`	inner diameter and length of vent 1; vent 2's diameter and telescope range
`vent_used`, `nf_mic_dist`	vent setting at the Zi/Nfield station; mic distance in the near field
`notes`	anything else (who did what, odd events, changes)

C MiniDSP filter texts

For reference only: **paste from the `.txt` files** in `Labs/Project/Final design/data/dsp/`, not from this PDF. The MiniDSP's a_1, a_2 have the opposite sign of the textbook form, as `DigitalCrossover.m` prints them ($a_1 = +1.9699$). Both biquads are identical in each file: two cascaded Butterworth sections ($Q = 0.707$) at 325 Hz, $f_s = 96$ kHz, make one LR4 filter.

Output 1, woofer:
`MiniDSP_woofer_out1.txt`

```
biquad1,
b0 = 0.000111435745339,
b1 = 0.000222871490678,
b2 = 0.000111435745339,
a1 = 1.969920198141118,
a2 = -0.970365941122475,
biquad2,
b0 = 0.000111435745339,
b1 = 0.000222871490678,
b2 = 0.000111435745339,
a1 = 1.969920198141118,
a2 = -0.970365941122475,
```

Output 2, mid + tweeter:
`MiniDSP_midtw_out2.txt`

```
biquad1,
b0 = 0.985071534815898,
b1 = -1.970143069631796,
b2 = 0.985071534815898,
a1 = 1.969920198141118,
a2 = -0.970365941122475,
biquad2,
b0 = 0.985071534815898,
b1 = -1.970143069631796,
b2 = 0.985071534815898,
a1 = 1.969920198141118,
a2 = -0.970365941122475,
```